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# JULIA B. NAKHLEH

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## EDUCATION

**University of Wisconsin-Madison**

Ph.D. in Computer Sciences; advisor: Robert D. Nowak

*September 2021 - June 2027 (expected)*

**University of Wisconsin-Madison**

M.A. in Mathematics

*September 2022 - May 2024*

**Arizona State University**

B.S. in Computer Science, B.A. in Spanish (Summa Cum Laude, GPA 4.0)

*August 2015 - May 2019*

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## RESEARCH INTERESTS

Mathematical foundations of deep learning; neural network function spaces; regularization, sparsity, and implicit bias; connections with harmonic analysis, nonparametric estimation, and compressed sensing.

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## PUBLICATIONS AND PREPRINTS

- [1] **Deep Neural Variation Spaces: A Unifying Perspective on Depth and Complexity**  
Julia B. Nakhleh, Robert D. Nowak  
*Preprint* [\[arXiv\]](#)
- [2] **Global Minimizers of  $\ell^p$ -Regularized Objectives Yield the Sparsest ReLU Networks**  
Julia B. Nakhleh, Robert D. Nowak  
*NeurIPS 2025* [\[proceedings\]](#) [\[arXiv\]](#)
- [3] **A New Neural Kernel Regime: The Inductive Bias of Multi-Task Learning**  
Julia B. Nakhleh, Joseph Shenouda, Robert D. Nowak  
*NeurIPS 2024* [\[proceedings\]](#) [\[arXiv\]](#)
- [4] **Training OOD Detectors in their Natural Habitats**  
Julian J. Katz-Samuels\*, Julia B. Nakhleh\*, Robert D. Nowak, Yixuan Li (\*equal contribution)  
*ICML 2022* [\[proceedings\]](#) [\[arXiv\]](#)
- [5] **Identifying Entangled Physics Relationships through Sparse Matrix Decomposition to Inform Plasma Fusion Design**  
M. Giselle Fernández-Godino, Michael J. Grosskopf, Julia B. Nakhleh, Brandon M. Wilson, John L. Kline, Gowri Srinivasan  
*IEEE Transactions on Plasma Science* 2021 [\[IEEEExplore\]](#) [\[arXiv\]](#)
- [6] **Exploring Sensitivity of ICF Outputs to Design Parameters in Experiments using Machine Learning**  
Julia B. Nakhleh, M. Giselle Fernández-Godino, Michael J. Grosskopf, Brandon M. Wilson, John L. Kline, Gowri Srinivasan  
*IEEE Transactions on Plasma Science* 2021 [\[IEEEExplore\]](#) [\[arXiv\]](#)

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## RESEARCH EXPERIENCE

**Graduate student researcher**, University of Wisconsin-Madison

*September 2021 - present*

- Developed a function-space theory of deep networks with general activations; derived novel structural results and complexity bounds [\[1\]](#).
- Proved sparsity guarantees and studied functional structure of shallow  $\ell^p$ -regularized ReLU networks [\[2\]](#).
- Characterized the implicit bias of multi-task ReLU networks and relationship to kernel methods [\[3\]](#).
- Earlier work developed algorithms for out-of-distribution detection with unlabeled auxiliary data [\[4\]](#).

**Post-baccalaureate researcher**, Los Alamos National Laboratory

*November 2019 - August 2021*

Applied sparse matrix decomposition and feature-importance methods to analyze physical relationships in nuclear fusion experiments. [\[5\]](#), [\[6\]](#).

**Undergraduate researcher**, Arizona State University (AAIR Lab)

*August 2018 - May 2019*

Studied the ability of neural networks to learn abstract state representations for classical planning problems.

**Undergraduate researcher**, Los Alamos National Laboratory

*May 2016 - August 2017*

Applied neural sparse coding methods to denoise, reconstruct, and classify Electroencephalography (EEG) signals.

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## SELECTED TALKS

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- **Global Minimizers of  $\ell^p$  Regularized Objectives Yield the Sparsest ReLU Networks**  
SIAM Conference on Optimization (OP26), minisymposium on Global Optimization for Neural Networks. Edinburgh, UK (2026).
- **Sparsest ReLU Neural Networks via  $\ell^p$  Regularization for  $0 < p \leq 1$**   
IFDS Workshop on Theoretical Foundations of Applied AI. Seattle, Washington (2025).
- **A New Neural Kernel Regime: the Inductive Bias of Multi-Task Learning**  
ONR MURI review (2025).

## TEACHING AND GRADING EXPERIENCE

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|                                                                                      |                    |
|--------------------------------------------------------------------------------------|--------------------|
| TA for ECE/CS 532 (Matrix Methods in Machine Learning), UW Madison                   | <i>Spring 2026</i> |
| Grader for CS/ECE/STAT 861 (Theoretical Foundations of Machine Learning), UW Madison | <i>Fall 2025</i>   |
| TA for MATH/STAT/ECE 888 (Nonparametric Methods in Data Science), UW Madison         | <i>Spring 2025</i> |
| Grader for MATH 718 (Randomized Numerical Linear Algebra), UW Madison                | <i>Fall 2024</i>   |
| Undergraduate TA for CSE 310 (Data Structures & Algorithms), ASU                     | <i>Fall 2018</i>   |
| Undergraduate TA for CSE 100 (Intro to C++), ASU                                     | <i>Spring 2018</i> |
| Math tutor (calculus I-III, linear algebra, discrete math, statistics), ASU          | <i>Fall 2016</i>   |

## ACADEMIC SERVICE

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Reviewer: JMLR (2025), NeurIPS (2025), IEEE Transactions on Information Theory (2025)

## AWARDS & SCHOLARSHIPS

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|------------------------------------------------------------------------------------------------|--------------------|
| SIAM Student Travel Award, OP26 conference                                                     | <i>2026</i>        |
| NSF IFDS Research Assistantship                                                                | <i>2025</i>        |
| NSF Graduate Research Fellowship Program (GRFP), Honorable Mention                             | <i>2021</i>        |
| ASME Student Presentation Award Winner, 2 <sup>nd</sup> place                                  | <i>2020</i>        |
| Phi Beta Kappa Honor Society                                                                   | <i>2019</i>        |
| Fulbright Scholarship (Spain - Study/Research), Semi-Finalist                                  | <i>2019</i>        |
| Marshall Scholarship, Finalist                                                                 | <i>2018</i>        |
| Undergraduate merit scholarships: National Merit Scholar; New American University Scholar, ASU | <i>2015 - 2019</i> |